

Individual Reflection on Group Project

Project Title: Graph-Based Cross-Modal Prediction from Spatial Transcriptomics in Glioma

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Team / Group: Group 13

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1. Your views on the completing the project as a team

Completing the project as a team required hard work, persistent and consistent contribution from each and every member to achieve our final results. To ensure the smooth completion of the project, we needed to divide the work into tasks that covered every aspect of the end-to-end data pipeline from pre-processing, to modelling, to result analysis and report writing. Each member worked on their individual pipelines and we routinely collaborated as a team to combine our results and discuss next steps.

2. Your views on project management and execution of your team

While the problem statement was precise, the possible methodologies to reach the outcome were vast and open ended. Given the time constraints, our team did a good job of identifying four state-of-the-art models and implementing them using the limited available resources. The division of labour was equal and fair which ensured a smooth and consistent execution of the tasks at hand.

3. Reflections on Successes, Failures, and Retrospective Changes

- **What Worked Well:** Assigning each team member to explore a different pipeline allowed for self-paced individual work within a predetermined time period while allowing the room for collaboration and team work to combine each other's approaches. As a team we worked well together with each member managing their time wisely, prioritising and completing the tasks as necessary.
- **What Did Not Work Well:** While the four different pipelines yielded good results, each model reached an early ceiling after spending a lot of time and resources on its implementation and training leaving little room to explore more models given the limited project time frame and available compute.
- **What I Would Do Differently:** I would personally try and explore as many baseline models as possible on a small subset of the data to ensure we have multiple possible approaches to solve the problem statement. I would then perform various analytical assessments to select the best methods and run them on the full data using AWS compute. In this way, we would try to avoid reaching a ceiling in terms of performance as well as manage our resources better

4. Any other reflections

The project highlighted the immense potential of data modelling in the field of medicine to help discover better treatment options for life threatening diseases like cancer. The project allowed us to expand our skills into the domain of bioinformatics through a real world problem. While data can be extremely helpful in biology, training these models comes with a significance cost both in terms of time and resources especially the compute required. This challenge taught us how to balance training time, AWS credit usage and model performance to deliver the most optimal result within these constraints.